



SOFTWARE DEV

MERN STACK DEVELOPMENT REPORT

Task Management Application – UI and Feature Enhancement

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LAB REPORT

Task: Task Management Application – UI and Feature Enhancement

1. Introduction

During this task, I worked on a **Task Management Application** as part of my practical software development work. The main purpose of this task was to improve the application’s user interface and implement additional functionality in the Notes module.

The work focused primarily on improving navigation, organizing notes through categories, and providing users with an easier way to filter and manage their notes.

The task was completed using a component-based mobile application structure. The implementation was carried out while maintaining the existing project structure and functionality.

2. Objectives

The main objectives of this task were:

- To improve the application’s navigation structure.
- To implement a bottom tab navigation system.
- To add categories to notes.
- To provide category selection while creating or editing notes.
- To implement category-based filtering.
- To improve the presentation of information inside note cards.
- To maintain a clean and user-friendly interface.
- To test the newly implemented functionality.

The objective was not to redesign the entire application but to enhance specific modules while keeping the existing functionality stable.

3. Technologies Used

The task was completed using modern mobile application development technologies.

Technology	Purpose
React Native	Mobile application development
JavaScript	Application logic
React Navigation	Screen and tab navigation
Local Storage	Local data persistence
Firebase	Data synchronization/backend services
Git	Version control
Visual Studio Code	Development environment

The exact internal backend structure and application-specific business logic are not included in this report.

4. Task 1 – Bottom Tab Navigation

The first major task was to implement and improve the application's bottom tab navigation.

Previously, navigating between different sections of an application can become inconvenient if the navigation structure is not properly organized. To improve accessibility, a bottom navigation structure was implemented.

The navigation provides users with quick access to the main sections of the application.

Main Work Performed

- Configured bottom tab navigation.
- Connected existing screens with the navigation structure.
- Added active-state highlighting.
- Improved navigation accessibility.
- Adjusted icon styling.
- Tested navigation between different screens.

The navigation structure was implemented in the application's main tab navigation component.

Result

After implementation, users can switch between major application sections directly from the bottom navigation area. The active tab is visually highlighted, making it easier to identify the current section.

5. Task 2 – Adding Categories to Notes

The second task was to introduce a category system for notes.

Without categories, managing a large number of notes can become difficult. Categories provide a simple way to organize notes based on their purpose.

For example, notes can be classified into categories such as:

- Work
- Personal
- Study
- Important

The category system was integrated into the existing note structure without changing the overall purpose of the Notes module.

Work Performed

- Added a category field to notes.
- Updated the note creation process.
- Updated the note editing process.
- Added category selection to the user interface.
- Ensured the selected category is associated with the relevant note.

This functionality was implemented in the screen responsible for creating and editing notes.

6. Task 3 – Category Filter Chips

After adding categories, the next task was to make the categories useful for managing a larger collection of notes.

For this purpose, category filter chips were implemented in the Notes screen.

The interface allows users to select a category and immediately view the notes belonging to that category.

Filtering Process

When a user selects a category:

1. The selected category is stored in the application state.
2. The notes collection is checked against the selected category.
3. Matching notes are placed in the filtered list.
4. The interface is updated automatically.
5. The user sees only the relevant notes.

The **All** option allows the user to return to the complete list of notes.

7. Task 4 – Note Card Enhancement

The final major UI task was to update the note card component.

The note card is responsible for presenting individual notes to the user. Since categories were introduced, the category information was also added to the note card.

The category is displayed alongside the existing note information.

Example

Complete Assignment

Finish the Software Engineering
assignment before submission.

Date: 11 August 2026

Category: Study

This small enhancement provides additional context without requiring the user to open the note.

Benefits

- Better information visibility.
 - Easier identification of note types.
 - Improved organization.
 - Better consistency with the new filtering functionality.
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8. Testing and Verification

After completing the implementation, the updated features were checked to ensure that they worked correctly with the existing application.

The following areas were considered during testing:

Test	Expected Result
Open application	Application loads normally
Bottom navigation	Tabs open correct screens
Active tab	Selected tab is highlighted
Create note	Note is created successfully
Select category	Category can be selected
Edit note	Category can be modified
Category filter	Relevant notes are displayed
All filter	All notes are displayed
Note card	Category is visible
Navigation switching	Screens change correctly

The testing focused on ensuring that the newly implemented features did not negatively affect the existing functionality.

9. Challenges Faced

During implementation, several development considerations were important.

Existing Code Integration

The new functionality had to be integrated into the existing project rather than being developed as an independent module.

State Management

The filtering functionality required the selected category and filtered notes to remain synchronized with the interface.

UI Consistency

The new category controls and navigation elements needed to match the application's existing visual structure.

Data Compatibility

Adding a new category field required consideration of existing notes that may not already contain category information.

These issues were addressed while maintaining the existing application's functionality.
